

MORE ABOUT SHAP

The SHAP framework, introduced by Lundberg and Lee at NeurIPS 2017, unifies several prior explanation methods under the game-theoretic framework of Shapley values. In cooperative game theory, the Shapley value of a player in a coalitional game is defined as the weighted average of the player's marginal contribution across all possible coalitions of players. This concept translates to machine learning by treating features as players and the model prediction as the game's payoff.

Formally, for a model f and an input instance x with features indexed 1 through p , the SHAP value for feature j is defined as the sum over all subsets S of the feature set excluding j , weighted by the inverse of the binomial coefficient, of the difference $f(S \cup j)$ minus $f(S)$, where $f(S)$ denotes the model's expected prediction when only the features in set S are known and all other features are marginalized out.

This definition satisfies three critical axioms. Local accuracy requires that the sum of all SHAP values equals the difference between the model's prediction for the given instance and the average prediction across the dataset. Missingness requires that features not present in the model receive zero attribution. Consistency requires that if a feature's marginal contribution increases in a modified model, its SHAP value does not decrease.

The computational challenge of exact Shapley value computation lies in the exponential number of feature subsets that must be evaluated. For p features, there are 2^p subsets, making exact computation infeasible for high-dimensional inputs. SHAP addresses this through model-specific efficient algorithms: TreeSHAP for tree-based models achieves polynomial-time exact computation by exploiting the tree structure, KernelSHAP for arbitrary models uses a weighted linear regression approximation, and DeepSHAP for deep learning models uses a connection between Shapley values and gradient-based attribution.

2.3 Alternative Explanation Methods

Several alternative explanation methods were considered during the design of the KeyPersona XAI module. LIME (Local Interpretable Model-agnostic Explanations) generates explanations by fitting an interpretable surrogate model to local perturbations of the input. While LIME is model-agnostic and computationally efficient, it lacks the theoretical guarantees of SHAP and can produce unstable explanations when the local perturbation distribution is poorly calibrated.

Permutation importance measures feature importance by quantifying the decrease in model performance when a feature's values are randomly permuted across the dataset. This approach provides a global importance ranking but does not provide instance-level explanations, limiting its utility for individual prediction explanation. Additionally, permutation importance can be misleading for correlated features, where permuting one feature disrupts the joint distribution and inflates the importance of correlated partners.

Gradient-based methods, including Integrated Gradients and Grad-CAM, compute feature attributions based on the partial derivatives of the model output with respect to the input features. These methods are computationally efficient for differentiable models but are not applicable to tree-based models such as random forests and gradient boosting, which are among the best-performing models in the KeyPersona pipeline.

The decision to adopt SHAP as the primary explanation framework was motivated by its theoretical rigour, its availability of model-specific efficient implementations, its applicability across all model families in the KeyPersona pipeline, and its provision of both global and local explanations within a unified framework.